MUHAMMAD DANISH

AI Engineer | Machine Learning Engineer | Generative AI Developer

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PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience building real-world AI solutions across Machine Learning, Deep Learning, Computer Vision, and Generative AI. Currently working on voice AI agents, computer-vision-based interior styling systems, and practical AI automation to solve real business and consumer problems. Experienced integrating AI models into production-ready applications using Python, TensorFlow, PyTorch, Scikit-learn, OpenCV, and FastAPI. Runner-Up at ByteSprint 2025 and an active participant in several national and international hackathons, with additional entrepreneurial experience running an e-commerce business.

EDUCATION

Bachelor of Computer Science — Specialization in Artificial Intelligence | Abdul Wali Khan University Mardan, Pakistan *Sept 2022 – 2026*

Courses: Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Data Structures & Algorithms, Data Engineering, Web Development, Database Systems, Software Engineering.

PROFESSIONAL EXPERIENCE

Junior AI/ML Engineer — Excels Tech Solution LLC *August 2026 – Present*

- Develop and integrate Generative AI applications using LLMs, RAG pipelines, AI agents, and vector databases for real-world business applications.
- Build and maintain backend AI services and APIs using Python and FastAPI, powering AI chatbot and customer-support automation with knowledge retrieval and prompt workflows.
- Contribute to multi-agent and MCP-based AI systems, integrating specialized tools and services into production workflows.
- Perform testing, debugging, QA, and hallucination analysis to improve the reliability and accuracy of AI-generated responses, collaborating with the team on architecture, deployment, and optimization.

Data Science Intern — 10Pearls, Pakistan *July 2026 – September 2026*

- Contributed as part of a team to Pearls AQI Predictor, an end-to-end machine learning pipeline for forecasting Air Quality Index (AQI) 3 days ahead using a serverless stack.
- Built feature engineering and model training components using Python, Scikit-learn, and TensorFlow, with features stored in Vertex AI's feature store.
- Set up automated pipeline runs using GitHub Actions for hourly feature updates and scheduled model retraining.
- Developed a real-time prediction dashboard using Streamlit and FastAPI to surface 3-day AQI forecasts to end users.

AI Engineer Intern — Codecilex, Pakistan *May 2025 – July 2025*

- Worked on computer vision and AI automation initiatives, integrating AI models into practical, production-ready applications.
- Collaborated with cross-functional teams to deliver production-ready AI solutions.

Text-to-Speech (TTS) Voice Agent *[Python | TTS Frameworks | FastAPI]*

- Developed a TTS voice agent that converts text input into natural, real-time speech output for automated speech applications.
- Designed the pipeline for conversational use cases, enabling automated, hands-free voice interaction.
- Integrated the agent via a FastAPI backend to support easy deployment into other applications.

AI Room Scan & Interior Styling System *[Python | OpenCV | Generative AI | FastAPI]*

- Built a computer-vision pipeline that scans a room image and identifies layout, furniture, and empty space.
- Combined the vision output with generative AI to produce personalized interior styling recommendations.
- Exposed the system through a FastAPI backend for integration into a practical, end-user-facing application.

PROJECTS

Excels Assistly — AI Customer Support, Live Chat & CRM *[FastAPI | Next.js 16 | PostgreSQL | RAG | OpenAI & Anthropic APIs | Cloud Run]* assistly.excelsnetworks.com

Excels Tech Solution LLC

- Built a multi-tenant SaaS where businesses handle customer chats, tickets and leads in one place, as a modular monolith (FastAPI backend with 18 modules, Next.js 16 frontend), delivered phase by phase from a written roadmap.
- Implemented an AI reply engine with a provider abstraction (OpenAI, Claude, local, offline test mode), per-workspace model choice, guardrails such as secret-leak blocking and output limits, and a safe fallback when a provider fails.
- Built RAG answers from each company's knowledge base: crawling, chunking, OpenAI embeddings with offline fallback, and a coverage rule deciding when semantic search is safe to use; plus an embeddable chat widget, real-time WebSocket messaging, shared inbox, tickets and CRM.

- Designed the superadmin console (organizations, plans, AI credit ledger, usage and cost, audit log) with strict tenant privacy; added 88 passing backend tests (pytest, Vitest, Playwright) and shipped on Cloud Run with Cloud Build CI/CD.

AI SEO — AI-native SEO Platform *[TypeScript | NestJS | Next.js 14 | PostgreSQL | Redis | GA4 | Claude / OpenAI / Gemini]* excelstechnology.com
Excels Tech Solution LLC

- Multi-tenant team SaaS where marketers and agencies crawl, audit and fix website SEO and measure brand visibility in Google and AI assistants.
- Built the Topical Authority module (topic breadth, depth, internal linking, pillar pages, AI-suggested missing subtopics) with scoring written as pure, unit-tested functions.
- Built AI Search Analytics, which reads Google Analytics 4 and classifies visits from 13 AI assistants (ChatGPT, Perplexity, Claude, Gemini, Copilot and others), plus the SXO and Voice, Entity & AI Mode modules with AI query fan-out checks.
- Implemented "API required" error handling (HTTP 424 plus UI banner) so the app never shows invented SEO data; integrated an Assistly support-ticket system and an AI RAG chatbot widget across all public pages.

Excels POS — Restaurant POS + KDS + ERP *[Multi-tenant SaaS | POS | KDS | ERP | AI Copilot]* restaraunt-pos.excelsnetworks.com
Excels Tech Solution LLC

- Built a multi-tenant restaurant platform that runs dine-in, takeaway, QR and delivery orders from one system, with KDS/KOT, live rider tracking, inventory and recipe costing, accounting, HRM and AI insights.
- Designed tenant isolation, a transaction pattern and a generic ResourceDef CRUD engine that auto-generates most CRUD screens from configuration.
- Solved production issues including DB pooler timeouts behind checkout 500 errors, a middleware matcher bug that bypassed route guards, race-free rider auto-assignment, and safe data cleanup with accounting reversals.
- Scale: 143 data models, 335 APIs, 188 pages and 773 commits.

Uraan.ai *[Python | FastAPI | Groq | RAG | LangChain | Supabase — 2nd Position, Comeback Pakistan]* www.uraan.app

- Built an AI-powered adaptive learning platform that helps students prepare for exams through personalized study assistance, AI-generated questions, and weak-area identification.
- Worked on the AI/LLM logic and backend, implementing a RAG pipeline with LangChain and Groq, and using Supabase for data storage via FastAPI endpoints.
- Selected from thousands of applicants for Comeback Pakistan, an 8-week venture-building program; achieved 2nd position after presenting Uraan.ai to a panel of 20+ founders, HR professionals, and investors.

AarogyaNet — Multi-Agent Healthcare Assistant *[Python | LLMs | LangChain | LangGraph | RAG | FastAPI | Streamlit]* app-wine-pi.vercel.app

- Designed a multi-agent healthcare assistant with specialized agents for symptom analysis, medical knowledge retrieval, and wellness recommendations.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline over a vector database to ground responses in trusted medical information.
- Built a coordinator agent to route requests and combine outputs from specialized agents into a single structured response.

Humrahi Hub *[Web Platform | Blockchain-Inspired Transparency — Hackathon Winning Project]* humrahihub.vercel.app

- Built a community-driven platform to solve the "Someone Paradox" by connecting verified community needs with donors and volunteers.
- Designed blockchain-inspired transparency and automated fund-flow tracking so every donation is visible and accountable.
- Won 1st Position at the 2nd AWKUM AGORA 2026 Hackathon as part of Team Tec-Hub.

Analyzing Student Behavior Before and After Classroom Sessions Using Deep Learning *[Python | TensorFlow/PyTorch | OpenCV | FER2013]*

- Built a computer-vision system to analyze students' facial emotions before and after classroom sessions using the FER2013 dataset.
- Compared pre- and post-lecture emotional states to measure engagement and attention trends for instructors.

AI-Social-Agent *[Python | LLMs | Generative AI | Prompt Engineering | LangChain | Streamlit]*

- Built an AI content assistant that generates and rewrites social media posts, captions, and hashtags from simple prompts.
- Applied prompt engineering to adapt content tone and formatting across LinkedIn, Instagram, and X.

AI Career System *[Python | Scikit-learn | Pandas | NumPy | Streamlit]*

- Built an AI-powered career recommendation platform that analyzes user skills, education, and interests to suggest suitable career paths.
- Implemented skill-gap analysis and personalized course/certification suggestions using classification and recommendation techniques.

Flood AI Chatbot *[Python | NLP | Prompt Engineering]*

- Built a conversational chatbot during the Flood.pk Hackathon (NIC Peshawar) to help users get flood-related information.
- Applied prompt engineering to keep responses accurate, relevant, and easy to understand for the general public.

Fake News Detection *[Python | Scikit-learn | NLP]*

- Trained an NLP-based text classifier to distinguish between reliable and unreliable news articles.

- Applied text preprocessing and feature extraction techniques to improve classification quality.

Heart Disease Predictor & Cancer Stage Prediction *[Python | Scikit-learn | Pandas | NumPy]*

- Built two separate ML classification models to predict heart disease risk and cancer stage from patient health data.
- Performed data cleaning, feature engineering, and model evaluation to improve prediction reliability.

Credit Risk Analysis & Email Spam Detection *[Python | Scikit-learn | Feature Engineering]*

- Developed a credit risk classification model to support financial risk scoring decisions.
- Built a separate email spam detection model using text-based feature engineering and classification techniques.

Movie Review & Financial News Sentiment Analysis *[Python | NLP | Scikit-learn]*

- Built an NLP sentiment classifier to analyze and label movie reviews as positive or negative.
- Built a second sentiment analysis model focused on financial news to gauge market-related sentiment.

Lung Disease Detection using Lung Sounds *[Python | TensorFlow/Keras | Deep Learning — University Project]*

- Applied audio signal processing to convert lung sound recordings into model-ready features.
- Trained a deep learning model to classify lung sounds and detect signs of lung disease.

TECHNICAL SKILLS

Programming Languages: Python, SQL, TypeScript
Machine Learning & Deep Learning: Scikit-Learn, TensorFlow, PyTorch, Neural Networks, Transfer Learning, Model Optimization
Generative AI & LLMs: LLMs, RAG Systems, MCP (Model Context Protocol), AI Agents, Agentic AI, LangChain, LangGraph, Prompt Engineering, Multi-Agent Orchestration, OpenAI API, Anthropic API, Groq
NLP & Transformers: Natural Language Processing, Hugging Face Transformers, Text Processing, Embeddings, Semantic Search
Computer Vision: YOLOv11, Object Detection, Image Segmentation, VGG16, CNN
Vector Databases & Search: Vector Databases, Vector Search, FAISS, ChromaDB, Qdrant, Hybrid Retrieval
Databases: PostgreSQL, Supabase
Backend & APIs: FastAPI, NestJS, REST APIs, WebSockets, API Integration, Streamlit, Gradio, Next.js
MLOps & Deployment: Docker, MLflow, CI/CD Pipelines, Google Cloud Run, Git, GitHub, Model Deployment
Data Analysis: Pandas, NumPy, Feature Engineering, Data Preprocessing, EDA

CERTIFICATIONS & TRAINING

- Machine Learning, AI & Data Science — eHunar
- Generative AI Application Developer — NCEAC HEC
- KPITB AI/ML Program
- Machine Learning — Coursera
- Python — Coursera
- Multiple additional Coursera certificates in AI, ML, and Python (6–8 certificates)

AWARDS & ACHIEVEMENTS

- **2nd Position** — Comeback Pakistan (Uraan.ai): selected from thousands of applicants for an 8-week venture-building program; presented to 20+ founders, HR professionals, and investors.
- **1st Position** — 2nd AWKUM AGORA 2026 Hackathon (Team Tec-Hub, Project: Humrahi Hub)
- **Runner-Up** — ByteSprint 2025 Speed Coding Competition
- Participant — NeoHacks GIKI Edition 2025 · Flood.pk Hackathon (NIC Peshawar) · IUPC 2026 · Hack Nation Global AI Hackathon · Studiio Hacks